ZCAS University

SCHOOL OF COMPUTING, TECHNOLOGY AND APPLIED SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

CIT 2641 – INTRODUCTION TO DATABASES SYSTEMS

DEFERRED EXAM

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Question one is Compulsory and must be attempted. The Question carries **40 Marks**
2. Answer Any **THREE** questions from the remaining set of questions.
3. Candidates must **not turn this page** until the invigilator tells them to do so.
4. Each Question Carries 20 Marks
5. You are required to answer each Question on a new page

THREE

Instructions: Answer Question ONE (Compulsory) and any other ~~TWO~~ Questions

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QUESTION ONE (40 MARKS)

a) Explain the differences between the notions of Database (DB), Database Management System (DBMS) and Database System (DBS). **(5 marks)**

b) Give an SQL statement to create a database named *school*. **(2 marks)**

c) Consider the school database below, where the primary keys are underlined:

student(student_id, name, address, gender)

course(code, course_name, year)

department(dept_number, date, department_name)

i) Write an SQL statement to display all records in the *staff* table. **(3 marks)**

ii) Give an SQL statement to add a new record into the *department* table. **(5 marks)**

iii) Write an SQL statement to delete all the records from the *course* table. **(3 marks)**

d) Discuss the result effect and possible correction of the SQL statement below: **(5 marks)**

```
UPDATE student SET name='HARON KIMATHI';
```

e) Define the following terms as used in Database Systems: **(4 marks)**

i) DDL

ii) DML

f) Outline three advantages of Database Systems. **(3 marks)**

g) Distinguish between a primary key and a foreign key as used in relational databases, giving an example of each. **(5 marks)**

h) Briefly explain the concept of data integrity and state any two types of integrity constraints used in database systems. **(5 marks)**

QUESTION TWO (20 MARKS)

a) Outline the major steps that you would take in setting up a database for a particular enterprise, with full explanation of an enterprise of your choice. **(10 marks)**

b) State and explain the three types of relationships in relational databases, using at least four pairs of database tables for each to depict the relationship. **(5 marks)**

c) Explain the role of a Database Administrator (DBA) in an organisation. **(5 marks)**

QUESTION THREE (20 MARKS)

a) Discuss any three application areas of database systems. **(6 marks)**

b) Describe the ANSI/SPARC three-level architecture for Database Management Systems software and explain the advantages it provides. **(9 marks)**

c) Distinguish between physical data independence and logical data independence as used in the ANSI/SPARC architecture. **(5 marks)**

QUESTION FOUR (20 MARKS)

a) Explain the differences between the following notions: **(6 marks)**

i) Database (DB)

ii) Database Management System (DBMS)

iii) Database System (DBS)

b) Explain the distinctions among the terms primary key, candidate key, and composite key. **(6 marks)**

c) Highlight the three Database Storage System categories. **(3 marks)**

d) With the aid of an example, explain what is meant by a *schema* in the context of database systems. **(5 marks)**

QUESTION FIVE (20 MARKS)

a) Discuss five various types of database systems. **(10 marks)**

b) Outline five components of a DBMS. **(5 marks)**

c) Briefly explain any two database models, giving an example of a DBMS product for each. **(5 marks)**

---- END OF EXAMINATION — BEST WISHES ----